

Electric test vehicle: data needed for the simulation

Please fill in or send what you have. Required items are needed before the simulation and training mean anything for this vehicle. Optional items have workable defaults and can be refined after the first flights. Units: mm, g, N, deg, s.

Required

What	Unit	Accuracy
☐ Total mass ready to fly, battery in	g	+10 g
☐ Mass of the fan / motor / ESC unit	g	+5 g
☐ Balance point (CG) measured from the top, battery in	mm	+5 mm
☐ Overall length and body diameter	mm	+2 mm
☐ Max static thrust at full throttle, charged battery (thrust stand)	N	+3 %
☐ Thrust at 25 / 50 / 75 % throttle (same test)	N	+3 %
☐ Thrust response: time to reach 63 % after a throttle step	ms	+20 ms
☐ Battery time at hover throttle	s	+10 s
☐ Thrust vectoring: max angle each axis, and pivot (or vane) position from the top	deg, mm	+0.5 deg, +5 mm
☐ Thrust vectoring servos: model, speed, PWM range at centre and limits	-, deg/s, us	datasheet
☐ Roll control: what it is (fans or wheel), max roll torque or fan thrust x arm	N m	+20 %
☐ Legs: number, foot circle diameter, feet below the bottom of the body	-, mm, mm	+5 mm
☐ Touchdown speed and tilt the legs survive	m/s, deg	drop test
☐ GPS: model, update rate (needed for the 10 m landing circle)	-, Hz	datasheet
☐ IMU and barometer: model, sample rate	-, Hz	datasheet
☐ Flight computer: board, control loop rate, language of the flight code	-, Hz, -	-
☐ Mission rules: height, hover time, landing radius, measured from where	m, s, m	confirm

Optional

What	Unit	Used until measured
☐ Pitch inertia (CAD, or a swing test)	kg m ²	from size and mass
☐ Roll inertia (CAD)	kg m ²	from size and mass
☐ Drag coefficient and centre of pressure (OpenRocket)	-, mm	0.8, mid-body
☐ Fan reaction torque: roll torque per newton of thrust	N m / N	0.003
☐ Servo delay and deadband	ms, deg	20 ms, 0.1 deg
☐ Sensor noise and bias: 30 s log with the board still on a table	csv	datasheet values
☐ GPS accuracy at the field: 2 min log standing still	csv	1 m
☐ Site elevation and expected wind on test day	m, m/s	300 m, 0-5 m/s

Also useful

A photo or CAD screenshot with the top, the CG and the thrust vectoring pivot marked. The raw thrust stand log. How the motor is cut in an emergency (kill switch, tether).